U 8

8.1

- 1 The results are almost the same as in our algorithm as we got 144/150, and the Weka Explorer got 143/150.
  Weka was much faster as it divides the data set in parts (10 folds). And chooses one of the folds to use for tests and the others for training. It does that 10 times changing the fold it uses so it tests every point.

- 2 Yes, we got the same results as before but with slightly smaller relative errors. Weka still run faster due to more efficient data structures and optimize run time.

8.2

IRIS.csv

Algorithm	Acc	F-Measure	Comment
Naive bayes	144 / 150 (96%)	96%	Works well with simple data
IBK (K=3)	143/150 (95,3%)	95,3%	To dis of a neighbour
Logistic regression	144/150 (96%)	96%	Linear boundry
J48	144/150 (96%)	96%	Over fitting is possible
Random Forest	143/150 (95.3%)	95.3%	Strong robustness
SMO	144/150 (96%)	96%	Good for this dataset

MNIST.csv (7000 instances)

Algorithm	Acc	F-Measure	Comment
IBK(3)	93,7%	93,7%	Good, but very heavy
Naive bayes	69,7%	69,3%	Too naive for this dataset
J48	81.7%	81.6%	It takes a lot of time
Random Forest	72.2%	72.2%	Very fast and good results
SMO	91.9%	91.9%	Good for this data set

The difference between accuracy and F-Measure is that accuracy only counts how many predictions are right. Whilst F-measure balances precision and recall, giving better insight into classifier performances per class.